

Phase 4H Printer Bridge Adapter Contract

This is a vendor-neutral adapter contract for a local utility bridge or network printer service. It does not implement a vendor driver. It defines the behavior IT/vendor teams should implement against the EMR API.

Goals

- Poll queued barcode print jobs.
- Deliver rendered `zpl`, `escpos`, or `html` payloads to a printer path.
- Acknowledge `printing`, `delivered`, `failed`, or `cancelled`.
- Preserve retry/fallback behavior from Phase 4F.
- Feed observability metrics from Phase 4G.

Authentication

- Use API bearer token configured for the clinic bridge account.
- Store token outside source code.
- Rotate token before pilot and after operator turnover.
- Bridge account should have only the permissions required for:

- `GET /api/inventory-barcode-print-jobs`
- `PATCH /api/inventory-barcode-print-jobs/:jobId/delivery`

Polling

```
GET /api/inventory-barcode-print-jobs?clinicId=<clinic-id>&deliveryStatus=queued&connectionType=utility_bridge
```

Recommended interval:

- normal: every 5 to 10 seconds
- degraded network: every 30 seconds
- after repeated failures: exponential backoff up to 2 minutes

The adapter must not delete jobs from the EMR.

Print Job Fields

Required fields to consume:

- `id`
- `printer_language`
- `connection_type`
- `delivery_status`
- `target_endpoint`
- `rendered_payload`
- `label_count`
- `delivery_attempt_count`
- `label_template_id`

- `requested_at`

Delivery Flow

1. Poll queued jobs.
2. Pick oldest job first.
3. Mark job as printing.
4. Send `rendered_payload` to printer.
5. Mark job as delivered or failed.

Mark printing:

```
PATCH /api/inventory-barcode-print-jobs/<job-id>/delivery
```

```
{
  "deliveryStatus": "printing"
}
```

Mark delivered:

```
{
  "deliveryStatus": "delivered"
}
```

Mark failed:

```
{
  "deliveryStatus": "failed",
  "deliveryError": "Printer offline"
}
```

Error Handling

The adapter should mark failed when:

- printer is offline
- target endpoint is unreachable
- payload language is unsupported
- paper/label media is missing
- driver returns an error
- print timeout is exceeded

The adapter should include concise `deliveryError` text that an operator can understand.

Retry And Fallback

The adapter should not create duplicate print jobs during failure recovery.

Recovery options:

- Operator uses browser/manual fallback with Phase 4F APIs.
- Operator retries after printer path is healthy.

- Adapter polls again after retry returns job to `queued` .

Observability

IT should review:

```
GET /api/reports/printer-bridge-health?clinicId=<clinic-id>&startDate=<date>&endDate=<date>
```

Minimum metrics to monitor:

- `queued jobs`
- `failed jobs`
- `fallback jobs`
- `retry jobs`
- `failed count by printer profile`

Local Adapter Logging

Each adapter event should log locally:

- `timestamp`
- `clinic ID`
- `print job ID`
- `printer profile or endpoint`
- `delivery status`
- `attempt count`
- `error text`

Do not log patient clinical content beyond the print job ID unless required for support and approved by the clinic owner.

Certification Checklist

- ☐ Polls `queued jobs` only
- ☐ Marks printing before delivery
- ☐ Marks delivered after successful print
- ☐ Marks failed with error text on failure
- ☐ Does not duplicate jobs during retry
- ☐ Supports configured payload language
- ☐ Handles printer offline
- ☐ Handles malformed endpoint
- ☐ Uses bearer token from secure config
- ☐ Logs job ID/status/error locally
- ☐ Operator can use fallback while adapter is down
- ☐ Printer health report shows activity after test